TP6 Hypothesis Testing - Group Assignment

Total points: 20 points

Exercise 1 [3 points]

A government statistician is evaluating whether recent economic policies have reduced the national unemployment rate. Last year, the unemployment rate was 8%, and new data collected this year show some improvement.

The statistician wants to test whether the true unemployment rate has decreased compared to last year's rate.

1. Formulate the null hypothesis (H_0) and the alternative hypothesis (H_1).
2. Which type of test should be used here (one-sided test or two-sided test)? Explain why.

Exercise 2 [3 points]

A psychologist measures the stress level of employees before and after a mindfulness training program. The goal is to test whether the training leads to a significant change in the average stress level.

1. Formulate the null hypothesis (H_0) and the alternative hypothesis (H_1) for testing whether the training has an effect on stress level.

   (*Hint*: you can use μ_{before} and μ_{after} to denote the true average stress level before and after the training program)
2. Describe what does "reject H_0" imply in this example.

Exercise 3 [4 points]

A school tests a new teaching method. In a sample of students, the mean exam score under the new method is 78, while under the traditional method it was 75.

After performing the hypothesis test, the result yields a p-value of 0.12 at a 5% significance level.

1. Formulate the null hypothesis (H_0) and the alternative hypothesis (H_1) for testing whether the new method leads to a different average score.

   (*Hint*: you can use μ_{diff} to denote the true mean difference in exam scores between the new and traditional teaching methods)
2. Which type of test should be used here (one-sided test or two-sided test)? Explain why.

3. Based on the test result, what conclusion should the school draw? Explain your answer.

Exercise 4 [5 points]

A data scientist is evaluating two machine learning models to determine which one performs better on a classification task. Model A (the baseline model) achieves 82% prediction accuracy, while Model B (a new model using additional features) achieves 85% accuracy on the same dataset.

To assess whether this observed improvement could be due to random variation in the sample, the data scientist performs a hypothesis test and obtains a p-value of 0.04 at a significance level of $\alpha = 0.05$.

1. Formulate the null hypothesis (H_0) and the alternative hypothesis (H_1) for testing whether Model B improves accuracy.

   (*Hint*: you can use μ_A and μ_B to denote the true accuracies of Model A and Model B.)
2. Based on the test result, should the data scientist claim that Model B is significantly better than Model A? Explain your answer.
3. If, instead of 0.05, the significance level α was set to 0.01 before examing the results, how would it affect the decision? Explain your answer.

Exercise 5 [5 points]

A web engineer claims that a new server-optimization technique reduces average page-loading time. A data analyst needs to investigate whether the optimization reduces average page-loading time. He measures the same 500 websites before and after deployment. The observed average loading time is slightly lower after optimization.

Let μ_{before} and μ_{after} denote the true mean loading times before and after optimization. A hypothesis test gives a p-value of 0.07 at a significance level of $\alpha = 0.05$.

> *"Because the observed average loading time was lower after optimization, we selected a one-sided test. We tested H_0: $\mu_{after} = \mu_{before}$ against H_1: $\mu_{after} < \mu_{before}$. The p-value of 0.07 means that there is a 7% probability that the optimization has no effect. Because $p > 0.05$, we accept H_0 and conclude that the optimization does not improve loading performance. Therefore, it should not be deployed."*

Identify and correct at least four distinct problems in the analyst's reasoning. For each problem, identify the problematic claim, explain the issue and provide a statistically defensible correction, using the following format.

Problematic statement:
Issue:
Correction:

Requirement for Moodle submission:

- Include the Group number and names of group members in the submitted file.
- Please don't copy the questions in your submitted file.
- Please submit your solutions to Exercise 1-5 as PDF by **Oct. 29th, 23:59**. Late submission will NOT be accepeted and will be automatically rejected.